DBM-SI Structural Intelligence Dictionary (v2)

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Repository: https://github.com/sizhet/Structural-Intelligence

Unified Terminology Across

- Digital Brain Model – Structural Intelligence (DBM-SI)
- CCC Mathematics
- CGCCC
- UTN
- Function Tunnel Intelligence (FTI)
- Autonomous Structural Intelligence (ASI)
- Task-Action Dual Calling Graphs (TAD-CG)
- Future Networked Intelligence (FNI)
- Human-AI Hybrid Civilization (HAHC)

Introduction

This document provides a unified reference for the terminology, abbreviations, architectural concepts, and algorithmic frameworks used throughout the Structural Intelligence ecosystem.

The purpose of this dictionary is not to impose rigid definitions, but to provide a common conceptual language across multiple research repositories and future developments.

As the Structural Intelligence stack continues to evolve, terminology may expand. However, the core principles remain centered on:

- Structure
- CCC Preservation
- Metric Navigation
- Runtime Evolution
- Collective Intelligence

PART I — Core Structural Intelligence

Abbreviation	Full Name	Description
DBM-SI	Digital Brain Model – Structural Intelligence	Structural intelligence framework emphasizing CCC, metric spaces, runtime evolution, and structural governance
SI	Structural Intelligence	Intelligence emerging from structural organization rather than purely statistical transformation
SSI	Structural System Intelligence	Intelligence arising from interacting structural systems
RSI	Recursive Structural Intelligence	Intelligence generated through recursive structural refinement
ASI	Autonomous Structural Intelligence	Structural systems capable of autonomous runtime evolution
HAHC	Human-AI Hybrid Civilization	Civilization-scale intelligence emerging from Human-AI collaboration
FNI	Future Networked Intelligence	Network-based collective intelligence emerging from interconnected knowledge systems
IR	Intermediate Representation	Structural representation layer between raw observations and intelligent operations
MET	Minimal Evolution Threshold	Smallest practical improvement capable of producing meaningful structural progress

PART II — CCC Mathematics
Common Concept Core Framework

Abbreviation	Full Name	Description
CCC	Common Concept Core	Shared structural identity across representations
Static CCC	Static Common Concept Core	CCC independent of behavior and time
Behavioral CCC	Behavioral Common Concept Core	CCC emerging through trajectories and actions
CCC Preservation	CCC Preservation Principle	Maintaining core structure during transformation
CCC Distance	CCC Relative Distance	Structural distance measured relative to CCC
CCC Attractor	CCC Attractor	Structural convergence point
CCC Field	CCC Attractor Field	Collection of interacting CCC attractors
CCC Retention Ratio	CCC Retention Ratio	Percentage of CCC preserved after transformation
CCC Drift	CCC Drift	Structural displacement of CCC across representations
CPA	Core-CCC-Preserved Analysis	Analytical framework preserving CCC during reasoning
CPG	CCC-Preserved Generation	Generative framework preserving CCC while allowing diversity

PART III — Metric Space Intelligence
Structural Geometry and Navigation

Abbreviation	Full Name	Description
MDT	Metric Differential Tree	Differential tree constructed within metric space
EDT	Euclidean Differential Tree	Differential tree constructed within Euclidean space
ADT	Active Differential Tree	Runtime adaptive differential tree
Metric Space	Metric Space	Geometry defined by distance relationships
Perspective Distance	Perspective Distance	Observer-dependent structural distance
Structural Distance	Structural Distance	Distance over structures rather than coordinates
Directional Metric	Directional Metric	Asymmetric structural distance
Metric Homing	Metric Homing	Navigation toward target regions
Metric Targeting	Metric Targeting	Structural search guided by distance relationships
Two-Phase Search	Two-Phase Search	Euclidean retrieval followed by metric reranking
MDT Root CCC	MDT Root CCC Approximation	Root node acting as local CCC approximation

PART IV — Event Intelligence and ELM
Event-Based Structural Modeling

Abbreviation	Full Name	Description
ELM	Event Language Model	Event-centered representation and prediction framework

Event Family	Event Family	Family of event-sequence representations of the same phenomenon
Family Center	Family Center	Collective center of an Event Family
Family Drift	Family Drift	Difference between original signal and family center
Noise Drift	Noise Drift	Drift caused by extraction errors or artifacts
Resolution Drift	Resolution Drift	Drift caused by granularity differences
Knowledge Drift	Knowledge Drift	Drift caused by newly captured information
Counter-Weight Curve	Counter-Weight Curve	Balancing curve used during Family Drift correction
Event Sequence	Event Sequence	Ordered structural event representation
Event Compression	Event Compression	Transformation from raw data into event structures

PART V — CGCCC and AI Coding
Structural Governance for Software Generation

Abbreviation	Full Name	Description
CG	Calling Graph	Structural graph of execution relationships
CGCCC	Calling Graph Common Concept Core	CCC extraction and governance framework for calling graphs
CGFL	Calling Graph Formal Language	Formal representation language for calling graphs
SOS	State-Operation-State	Canonical structural runtime representation

TMI	Triple Mutual Inference	SO→S, SS→O, and OS→S inference framework
Vertical Gap Bridging	Vertical Gap Bridging	Bridging architectural intent to implementation details
Horizontal Gap Bridging	Horizontal Gap Bridging	Bridging incomplete state structures
Evidence Fiber	Evidence Fiber	Provenance-preserved path through knowledge structures
Feasible Path	Feasible Path	CCC-consistent transformation path
Compliance Surface	Compliance Surface	Accept/Reject/Rewrite decision space
Certified Path Reuse	Certified Path Reuse	Structural reuse scoring mechanism

PART VI — Universal Typing and Naming Structure-First Naming Systems

Abbreviation	Full Name	Description
UTN	Universal Typing and Naming	Structural naming and typing framework
Canonical Name	Canonical Name	Stable structure-derived identifier
CCC Naming	CCC-Based Naming	Naming derived from Common Concept Core
Reverse Naming	Reverse Naming	Inferring names from structure
Naming Alignment	Naming Alignment	Structural alignment across representations
Type Core	Type Core	Structural identity of a type
Naming Stability	Naming Stability	Persistence of naming under transformation

PART VII — Function Tunnel Intelligence (FTI)

Feasible Evolution and Navigation

Abbreviation	Full Name	Description
FTI	Function Tunnel Intelligence	Intelligence through navigation inside feasible functional tunnels
Functional Tunnel	Functional Tunnel	CCC-preserved feasible evolution corridor
Tunnel Navigation	Tunnel Navigation	Search and movement within feasible regions
Tunnel Family	Tunnel Family	Collection of related feasible tunnels
Tunnel Center	Tunnel Center	Representative trajectory of a tunnel family
CPA	Core-CCC-Preserved Analysis	Tunnel-preserving analysis framework
CPG	CCC-Preserved Generation	Tunnel-preserving generation framework
Functional Feasibility	Functional Feasibility	Ability to remain inside a valid tunnel
Tunnel Drift	Tunnel Drift	Deviation from the tunnel center

PART VIII — Autonomous Structural Intelligence (ASI)

Runtime Structural Evolution

Abbreviation	Full Name	Description
ASI	Autonomous Structural Intelligence	Autonomous structural runtime evolution framework
Elastic Trajectory	Elastic Trajectory	Structure-preserving adaptive path
Elastic Merge	Elastic Merge	Controlled trajectory fusion
Runtime Governance	Runtime Governance	Runtime structural control mechanisms

Structural Controller	Structural Controller	Runtime decision-making component
Trajectory Intelligence	Trajectory Intelligence	Intelligence operating on paths and trajectories
Go-Around	Go-Around Strategy	Constraint-preserving obstacle bypass
Structural Runtime	Structural Runtime	Runtime composed of evolving structural entities

PART IX — Task-Action Dual Calling Graphs
Task-Centric Structural Growth

Abbreviation	Full Name	Description
TAD-CG	Task-Action Dual Calling Graph	Coupled task/action structural representation
TG	Task Graph	Goal-space graph
AG	Action Graph	Execution-space graph
Dual Calling Graph	Dual Calling Graph	Integrated task and action representation
Delta Growth	Delta Growth Model	Growth through graph deltas
TGGG	Task Graph Growth Game	Experimental framework for task-graph evolution
MET-FTI Selection	MET-FTI Selection	Tunnel-guided task selection mechanism
Task Mutation	Task Mutation	Structural evolution of task graphs

PART X — Future Networked Intelligence (FNI)
Collective Knowledge Evolution

Abbreviation	Full Name	Description

FNI	Future Networked Intelligence	Collective intelligence emerging from networked knowledge systems
Family Dynamics	Family Dynamics	Evolution of Event Families
Network Dynamics	Network Dynamics	Evolution of knowledge networks
Flooding Zone	Flooding Zone	Region traversed by many function paths
Family Drift	Family Drift	Collective displacement of a representation family
Knowledge Fertility	Knowledge Fertility	Ability of a region to support future evolution
Flood Coverage	Flood Coverage	Fraction of paths traversing a region
Family Core Node	Family Core Node	High-coverage node within a knowledge family
Collective Intelligence	Collective Intelligence	Intelligence emerging from many agents
Civilization Drift	Civilization Drift	Long-term evolution of collective knowledge

PART XI — Human-AI Hybrid Civilization Civilization-Scale Intelligence

Abbreviation	Full Name	Description
HAHC	Human-AI Hybrid Civilization	Human-AI co-evolution toward civilization-scale intelligence
Human-AI Loop	Human-AI Loop	Continuous feedback between humans and AI
Participation Economy	Participation Economy	Civilization model based on contribution and collaboration

Knowledge Civilization	Knowledge Civilization	Civilization driven by structured knowledge creation
Networked Intelligence	Networked Intelligence	Intelligence distributed across many interconnected nodes
Collective Discovery	Collective Discovery	Knowledge generation through large-scale collaboration

PART XII — Canonical One-Line Definitions

CCC
The smallest structure that preserves identity.

UTN
Structure-first naming and typing.

CGCCC
Governance of software generation through Common Concept Core preservation.

ELM
Understanding the world through event structures rather than raw observations.

FTI
Navigation through feasible functional tunnels.

ASI
Autonomous evolution of structural runtime systems.

TAD-CG
Growth and execution through coupled task-action structures.

FNI
Collective intelligence emerging from networked knowledge evolution.

HAHC

Human-AI co-evolution toward civilization-scale intelligence.

DBM-SI

A structural intelligence paradigm built upon CCC, metric spaces, runtime evolution, and collective knowledge systems.

Final Orientation

Traditional AI often follows:

Input
↓
Universal Model
↓
Output

Structural Intelligence instead investigates:

Observation
↓
IR
↓
CCC
↓
Metric Navigation
↓
Structural Runtime
↓
Evolution
↓
Collective Intelligence
↓
Civilization Growth

The long-term vision of the Structural Intelligence ecosystem is to understand how intelligence emerges, evolves, and scales through structure-preserving processes across individuals, systems, and civilizations.